Investment Thesis Guy Jansen 29-05-2025

ACM Research, Inc.

[ACMR:NDAQ] Semiconductor Front End Processing Equipment

Price:	$ 22,59	Mkt Cap (M):	$ 1,444.14
Shares outstanding:	57,606,036	Enterprise Value (M):	$ 1,425.54
Beta (3Y Adj.):	1.42	Dividend Yield:	0,0 %
WACC:	9,73	P/E:	14,42

Executive Summary

ACM Research, Inc. (ACMR) presents a compelling long investment opportunity, driven by its differentiated semiconductor cleaning and advanced processing technologies, strong leverage to China's secular semiconductor self-sufficiency drive, and a rapidly expanding addressable market through new product introductions. While facing geopolitical headwinds and customer concentration risks, ACMR's robust growth (FY25 revenue guidance: 850M-950M, ~15% YoY growth at midpoint), improving operational efficiency, and strategic global diversification initiatives (e.g., Oregon facility) position it for sustained outperformance.

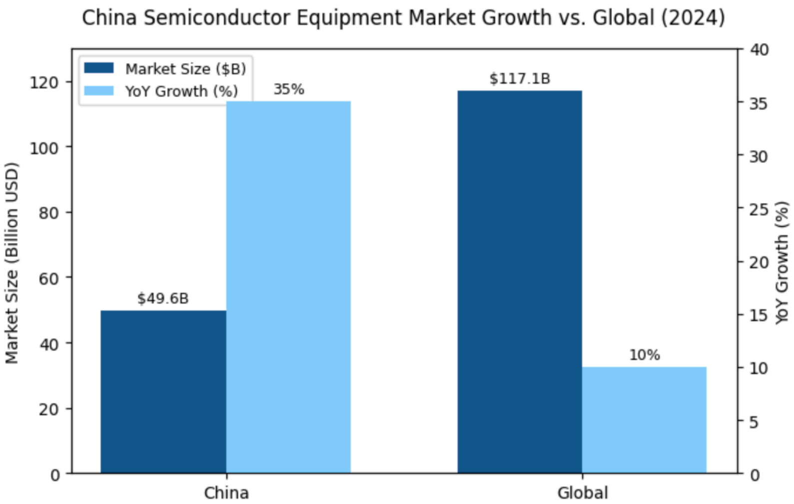
My analysis indicates an intrinsic value significantly above the current market price. The company's leadership in specialized cleaning, successful entry into new high-growth segments like ECP (including panel-level plating), Furnaces, and emerging Track/PECVD tools, supports a strong multi-year growth outlook. Despite trading at a historical discount to global peers, I believe continued execution, particularly on new product ramps and initial global customer traction, will serve as key catalysts for a valuation re-rating.

Recommendation:	**OVERWEIGHT / BUY**
Price Target (12-18 Months):	**$33.37**
Current Price (approx.):	$22.58
Implied upside:	47.8%

1. Investment Thesis

My positive stance on ACMR is predicated on the following key pillars:

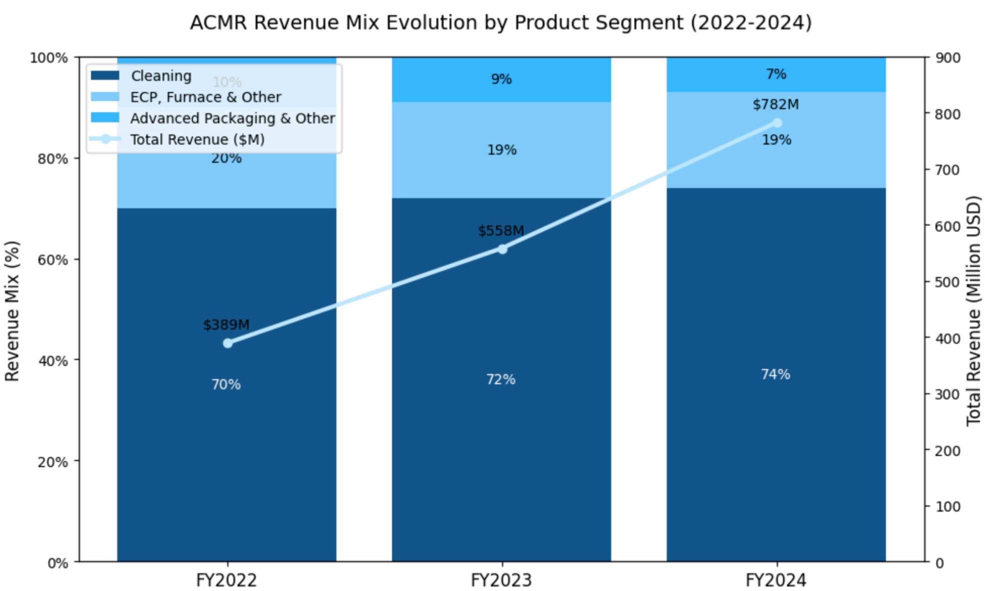
Dominant Niche & Secular Tailwinds in China: ACMR is a primary beneficiary of China's multi-billion dollar push for semiconductor self-sufficiency. Its specialized cleaning technologies are critical for advanced node manufacturing, where Chinese fabs are aggressively investing. This provides a strong, visible demand runway.



Differentiated Technology & Expanding Product Portfolio:

- **Cleaning Leadership:** Proprietary SAPS and TEBO technologies offer superior particle removal and lower cost-of-ownership. The recently qualified high-temperature SPM tool by a key logic customer underscores its technical edge.

- **SAM Expansion:** Successful diversification beyond cleaning into ECP (award-winning Ultra ECP ap-p for panel-level plating), Furnaces (significant ramp expected 2025), and emerging Track/PECVD platforms dramatically increases ACMR's total addressable market (estimated 2024 SAM of $18B).



Strategic Global Diversification & Capacity Expansion:

- The new Lingang facility significantly boosts production capacity to support growth.
- The Oregon R&D and initial production facility is a crucial step in de-risking China concentration, accessing global talent, and serving U.S./European customers directly.

Strong Financial Execution & Improving Profitability:

- Consistent >40% revenue CAGR over the past several years (FY24: +40.2%).
- Q1 2025 revenue of $172.3M (+13.2% YoY).
- Non-GAAP Gross Margins consistently within or exceeding the 42-48% target range (Q1 2025: 48.2%).
- Healthy balance sheet with net cash of $271M (Q1 2025) providing financial flexibility.

Valuation Dislocation:
ACMR historically trades at a significant discount to its global WFE peers. While some discount is warranted due to geopolitical risk and scale, I believe the current gap undervalues its growth potential, technological leadership in niche areas, and diversification efforts.

---

2. Company Overview

Founded in 1998 by Dr. David H. Wang, ACM Research designs, develops, manufactures, and sells single-wafer wet cleaning equipment and other advanced processing tools essential for semiconductor manufacturing. Headquartered in Fremont, California, its primary R&D, manufacturing, and sales operations are centered in Shanghai, China, through its subsidiary ACM Shanghai. This unique U.S.-domiciled, China-focused operational model has allowed it to deeply penetrate the rapidly growing Chinese semiconductor market.

Key Operational Highlights (Q1 2025):

- Single-wafer high-temperature SPM tool qualified by a key logic device manufacturer in mainland China.
- Ultra ECP ap-p tool won the 2025 3D InCites Technology Enablement Award for innovation in horizontal plating for panel-level packaging.
- Lingang production and R&D center (Shanghai area) is nearly completed.
- Oregon facility (purchased Oct 2024, 39,500 sq. ft.) is laying groundwork for initial production capacity and U.S. R&D/demonstration.
- Appointment of Charlie Pappis to Board of Directors.

3. Product Portfolio & Technology

ACMR’s portfolio targets critical steps in semiconductor fabrication and packaging:

Single Wafer Cleaning (Core Strength):

- **Products:** SAPS, TEBO, Tahoe, SPM (Sulfuric Peroxide Mix) tools.
- **Technology:** Proprietary megasonic cleaning (SAPS/TEBO) provides highly uniform and damage-free cleaning crucial for advanced nodes (<10nm). Tahoe technology significantly reduces chemical consumption. High-temperature SPM tools address critical resist stripping and post-etch cleaning.
- **Best Performing:** SAPS and TEBO platforms remain the cornerstone, with recent qualification of the high-temperature SPM tool by a leading Chinese logic fab highlighting continued innovation and customer adoption.

ECP (Electrochemical Plating) Systems:

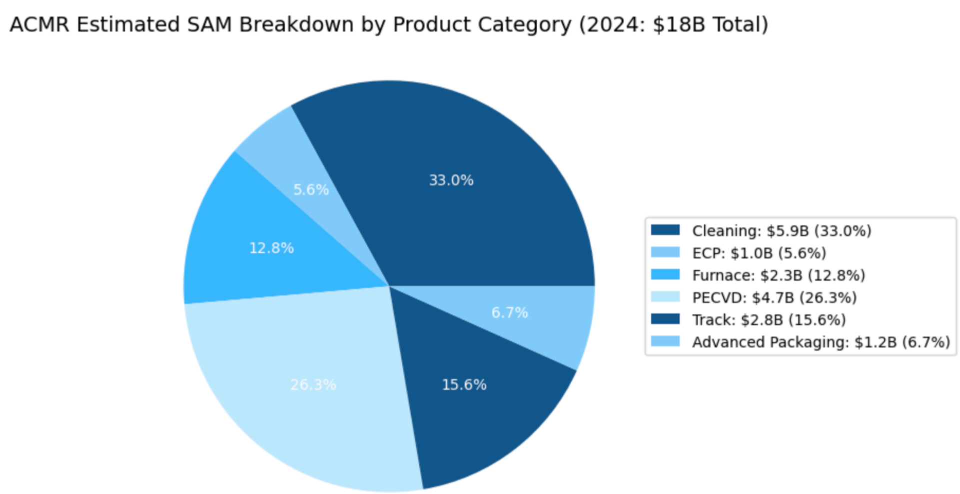
- **Products:** Ultra ECP AP (Advanced Packaging), Ultra ECP MAP (Front-End), Ultra ECP TSV, and the award-winning Ultra ECP ap-p (panel-level packaging).
- **Technology:** Delivers high-uniformity copper, nickel, and tin-silver plating. The Ultra ECP ap-p utilizes a novel horizontal plating approach for large-panel applications, crucial for next-gen AI chip packaging.
- **Best Performing:** The ECP segment, particularly for advanced packaging applications, is showing strong momentum. The Ultra ECP ap-p is a key differentiator and growth driver.

Furnace & Other Thermal Processes:

- **Products:** Ultra Fn A (Anneal), Ultra Fn O (Oxidation), Ultra Fn L (LPCVD), Ultra Fn P (PEALD).
- **Technology:** Vertical furnaces capable of high temperatures (up to 1250°C) for various deposition and thermal treatment steps. PEALD tools cater to advanced logic and memory.
- **Best Performing:** The overall Furnace segment is gaining significant traction, with strong customer evaluations and expected to be a material revenue contributor from 2025.

Emerging Platforms (Track & PECVD):

- **Products:** Inline KrF Track (Coater/Developer) systems and PECVD tools.
- **Technology:** Leveraging ACMR's platform innovation for high throughput and process flexibility in these large, established market segments.
- **Status:** In customer evaluation, initial revenue expected in 2025, ramping in 2026+.



4. Market Overview & Competitive Landscape

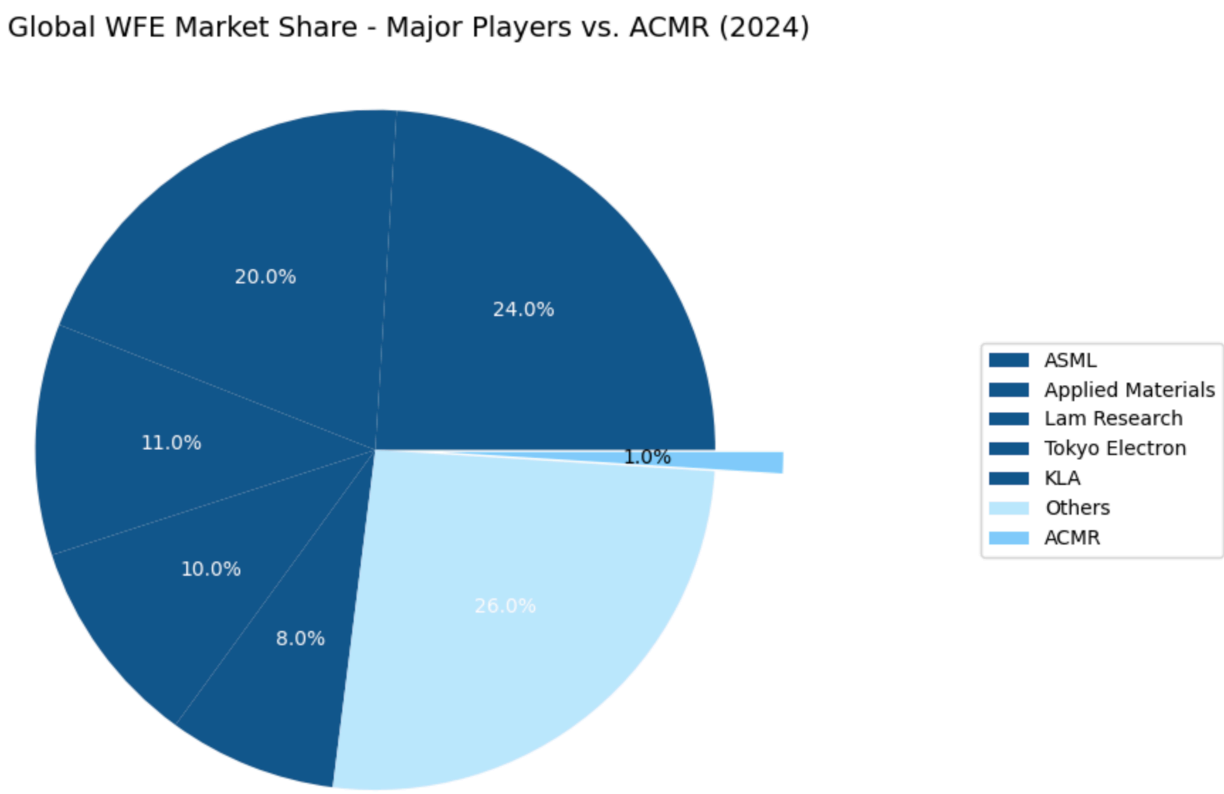
Market Drivers:

- **China's Semiconductor Self-Sufficiency:** The primary driver for ACMR. China aims to significantly increase domestic chip production, leading to massive fab construction and equipment demand.
- **Global Semiconductor Growth:** Driven by AI, 5G, IoT, automotive, and data centers. While ACMR is China-focused, this underpins overall industry health.
- **Advanced Packaging:** Heterogeneous integration and panel-level packaging are becoming critical for high-performance computing, creating demand for specialized ECP tools.
- **Technology Transitions:** Shrinking nodes require more advanced and precise cleaning and deposition steps, favoring companies with differentiated technology.

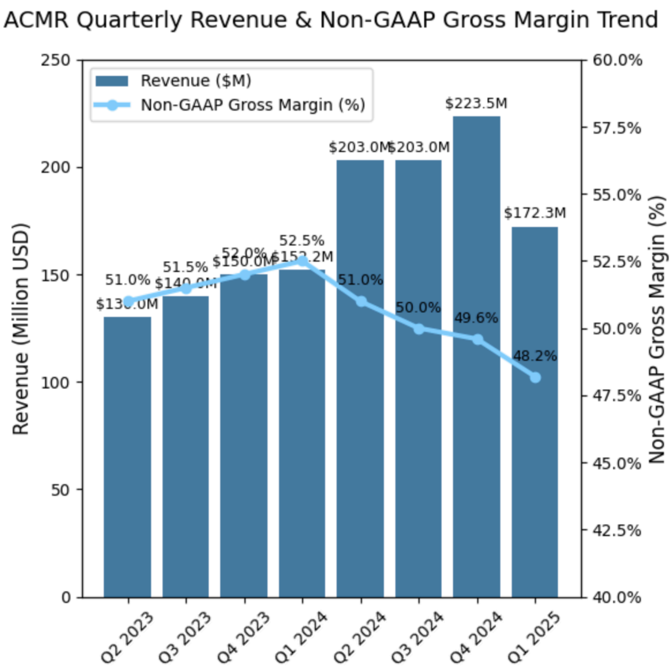
Competitive Landscape:

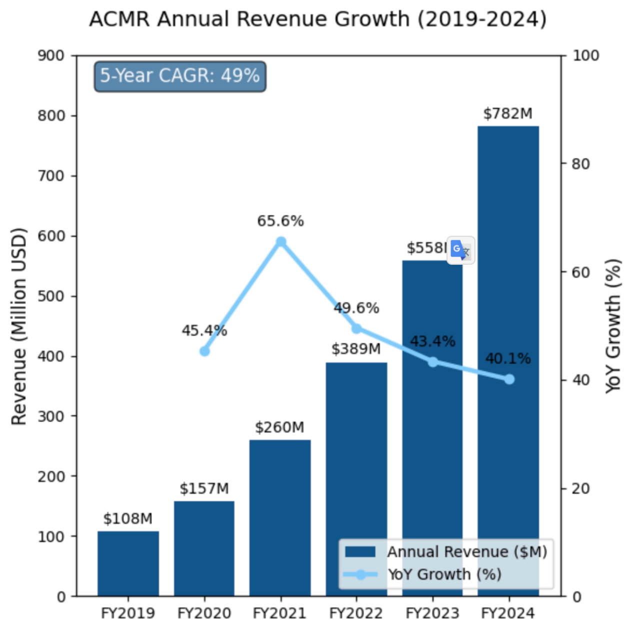
- **Global WFE Leaders:** Lam Research, Tokyo Electron (TEL), Applied Materials (AMAT), SCREEN. These are significantly larger, with broader portfolios and global reach.
- **ACMR's Niche:** Strong in specific cleaning applications in China due to technology and local presence. Expanding as a challenger in ECP, Furnace, and soon Track/PECVD.
- **Barriers to Entry:** Extremely high due to R&D costs, IP, long customer qualification cycles, and established relationships.

- **ACMR's Advantages:**
  - Technological differentiation (SAPS, TEBO, panel-level ECP).
  - Strong local engineering and support in China.
  - Agility and customer responsiveness.
- **ACMR's Challenges:**
  - Scale and resources compared to giants.
  - Geopolitical risks impacting access to certain technologies or markets.
  - Customer concentration.



5. Financial Analysis & Projections





Historical Performance & Key Metrics (Q1 2025 as Latest)

Metric	Q1 2025	Q1 2024	FY 2024	FY 2023	Trend / Notes
Revenue	$172.3M	$152.2M	$782.1M	$557.7M	Strong YoY Growth (+13.2% for Q1)
Shipments	$157.0M	$245.0M	$973.0M	$596.6M	Q1 '25 down YoY due to tough comp/pull-ins
Non-GAAP Gross Margin	48.2%	52.5%	50.4%	49.8%	Consistently strong, above 42–48% target range
Non-GAAP Operating Income	$35.6M	$39.8M	$200.6M	$123.2M	Q1 '25 down YoY, but strong absolute level
Non-GAAP Operating Margin	20.7%	26.2%	25.6%	22.1%	Healthy margins
Non-GAAP Diluted EPS	$0.46	$0.52	$2.26	$1.63	Reflects operational performance
Cash & ST Deposits	$498.4M	$441.9M (YE24)	$441.9M	$369.1M (YE23)	Strong liquidity
Net Cash	$271.0M	$259.0M (YE24)	$259.0M	$248.3M (YE23)	Increasing net cash position
R&D % of Revenue (Non-GAAP)	14.3%	12.8%	11.7%	15.1%	Sustained investment in innovation

Table 1: Historical Financial Metrics Summary

FY 2025 Outlook (Company Guidance):

- **Revenue:** $850 million - $950 million (implying ~15% YoY growth at midpoint).
- **Capex:** Expected ~$70 million.
- Effective Tax Rate: 10-15%.

Key Financial Strengths:

- High historical and projected revenue growth.
- Strong gross margin profile.
- Healthy balance sheet with growing net cash.
- Positive operating cash flow generation ($5.3M in Q1 2025).

Areas for Monitoring:

- Shipment linearity and conversion to revenue.
- Impact of product mix on gross margins (expected to stay in 42-48% range).
- Operating expense discipline as the company scales and invests in global expansion.

---

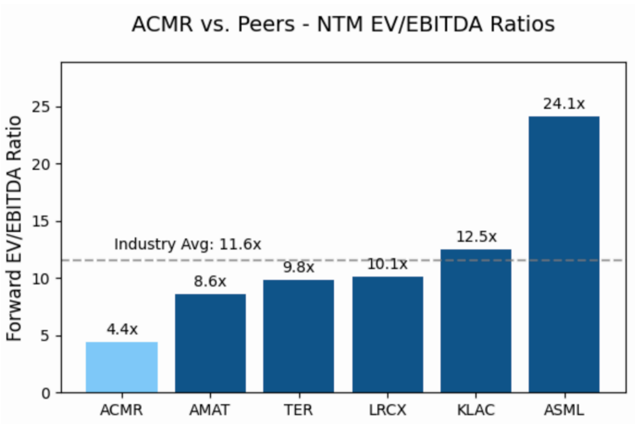
6. Valuation

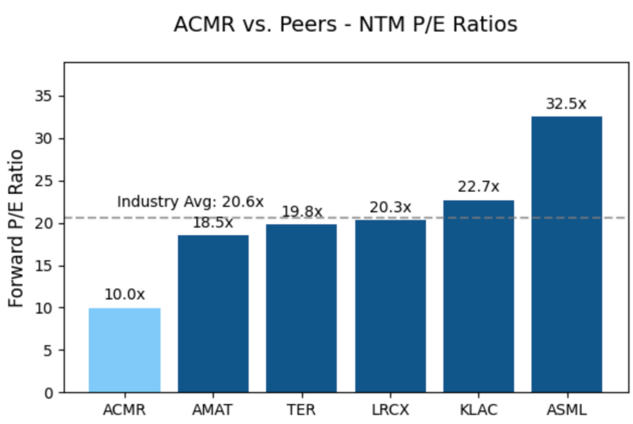
ACMR currently trades at a notable discount to its larger, more diversified global WFE peers. Comparable Company Analysis (Illustrative, based on prior analysis & market conditions):

Comparable Company Analysis (Illustrative, Based on Prior Analysis & Market Conditions)

Table 2: Comparable Company Valuation Multiples

Metric	ACMR (NTM Est.)	Peer Median (NTM Est.)
P/E	~9–11x	~18–22x
EV/Sales	~1.5–2.0x	~4.0–5.0x
EV/EBITDA	~6–8x	~12–15x





Valuation Rationale & Implied Upside:

- **Discount Factors:** Geopolitical risk associated with China concentration, smaller scale, less diversified product offering compared to giants, ADR structure.
- Potential for Re-rating:
  1. Continued strong execution and market share gains in China.
  2. Successful commercialization and ramp of new products (Furnace, Track, PECVD), proving SAM expansion.
  3. Tangible progress and revenue from global diversification (Oregon facility, non-China customer wins).
  4. Sustained margin strength.
- The prior blended DCF-CCA valuation (pre-Q1 '25 update) suggested an intrinsic value of **$33.37**, implying significant upside. The Q1 2025 results and maintained guidance reinforce the underlying strength.

7. Key Investment Drivers & Catalysts

Short-to-Medium Term (6-18 months):

- **New Product Revenue Ramp:** Significant revenue contribution from the Furnace product line in 2025. Initial revenues from Track and PECVD tools.
- **Customer Qualifications:** Further qualifications of advanced cleaning tools, ECP, and Furnace products at new and existing customers, both in China and globally.
- **Lingang Facility Operationalization:** Full ramp-up of the Lingang facility, enhancing production capacity and R&D capabilities.
- **Oregon Facility Progress:** Milestones in establishing the Oregon facility (demo capabilities, initial production setup).
- **Sustained Margin Performance:** Demonstrating ability to maintain gross margins within or above the high end of the 42-48% target range.
- **Geopolitical Stability/Clarity:** Any positive developments or increased clarity on US-China trade relations or export controls could reduce perceived risk.

Long-Term (2-5 years):

- **China's Self-Sufficiency Trajectory:** Continued strong domestic investment in China's semiconductor industry.
- **Successful SAM Expansion:** Establishing meaningful market share in Track, PECVD, and other new product segments.
- **Global Customer Diversification:** Securing significant orders and building a sustainable revenue stream from customers outside mainland China.
- **Technology Leadership:** Maintaining an innovative edge in core and new product areas through sustained R&D.
- **M&A / Strategic Partnerships:** Potential for inorganic growth or partnerships to accelerate market access or technology acquisition (speculative).

8. Risk Factors & Risk reducers

Geopolitical Risk & Export Controls:

- **Risk:** Escalation of US-China tensions leading to broader/stricter export controls, impacting ACMR's sales to China or its ability to source key components. Potential for Chinese retaliatory measures.
- **Risk reducer:** US incorporation, strategic sourcing, focus on non-US components where feasible, development of Oregon facility for global supply chain, active engagement with regulatory bodies.

Customer Concentration & China Market Dependency:

- **Risk:** Over-reliance on a few large Chinese customers (HLMC, SMIC, YMTC, etc.). A slowdown at any key customer could materially impact ACMR.
- **Risk reducer:** Actively expanding customer base within China to Tier 2/3 fabs, strategic push for global customer wins, product diversification to appeal to a wider customer set.

Competitive Pressure:

- **Risk:** Intense competition from larger, well-resourced global players and emerging local Chinese competitors. Potential for price erosion and margin pressure.
- **Risk reducer:** Focus on differentiated technology and IP, superior performance in niche applications, strong local customer support in China, continuous innovation.

Industry Cyclicality:

- **Risk:** The semiconductor equipment industry is inherently cyclical. A downturn in global semi capex could impact ACMR’s growth.
- **Risk reducer:** China's secular self-sufficiency drive may partially decouple its spending from global cycles. Strong balance sheet provides resilience.

Execution Risk:

- **Risk:** Delays in new product development/qualification (Track, PECVD), challenges in ramping new facilities (Lingang, Oregon), or quality control issues.
- **Risk reducer:** Experienced management team, phased product rollouts, significant investment in R&D and manufacturing infrastructure.

Supply Chain Constraints:

- **Risk:** Dependence on specific suppliers for critical components; potential disruptions.
- **Risk reducer:** Dual sourcing where possible, strategic inventory management, strong supplier relationships.

---

9. Conclusion & Recommendation

ACM Research stands out as a high-growth, technologically differentiated player in the semiconductor equipment market, with a unique and strong position to capitalize on China's long-term semiconductor ambitions. The company is successfully expanding its product portfolio into large adjacent markets, significantly increasing its SAM. Recent strategic moves to establish a U.S. operational base in Oregon signal a proactive approach to global diversification and risk mitigation. While geopolitical tensions and customer concentration present undeniable risks, ACMR’s consistent financial outperformance, robust technological roadmap, and clear strategy for addressing these challenges are encouraging. The current valuation appears to offer an attractive entry point for investors with a 3-5 year horizon, providing exposure to secular growth trends at a discount to peers.

I reiterate my **OVERWEIGHT / BUY** recommendation on ACM Research (ACMR).